Scaling Critical Infrastructure While Reducing Resource Consumption Using AWS with Stockholm Exergi

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“We’ve reduced our carbon foot|
CO2 equivalent to less than 4 kç
Manager, Stockholm Exergi

With great power comes great responsibility. Stockholm Exergi, an energy provider responsible for powering heat to 800,000 people, takes that commitment to heart. With critical infrastructure and a commitment to sustainability, the company sought a way to reduce energy consumption and optimize the efficiency of its district's buildings.

In 2016, it began the development of its solution, Intelligy Solutions, on Amazon Web Services (AWS) to do just that. Now, customers in over 10,000 connected buildings across Stockholm benefit from living in energy-efficient, "smart" buildings.

Optimizing Energy Use and Reducing Emissions Using Intelligy Solutions on AWS

With Sweden's energy consumption expected to rise by as much as 150 percent by 2045, Stockholm Exergi recognized the value in shifting from a power plant–centric focus to a people-centric one. To gain insights into day-to-day energy use, it installed edge nodes into customer substations. The installed hardware facilitated the company's ability to control power use at those delivery points.

In conjunction with the hardware rollout, the company also began a large-scale software development project on AWS to build its custom solution, Intelligy Solutions, which was designed to specifically address the unique needs of its customers. Stockholm Exergi uses Amazon Simple Storage Service (Amazon S3), an object storage service offering exceptional scalability, to store its data. For its connected devices, it uses [illegible]e, which makes it simple to connect billions of Internet of Things (IoT) devices and route trillions of messages to AWS services without managing infrastructure.

By digitalizing the district’s heating and cooling networks, the company is better positioned to mitigate risk. “If we ever have a situation in which we can’t cover the full needs of Stockholm, we can adjust the levels of heat outtake to optimize resource distribution,” says Patrik Höjner, head of IoT solutions at Stockholm Exergi.

What’s more, using AWS, the company found a way to take advantage of slack in its system to shave energy use during peak consumption times. For example, in the mornings, when a lot of people shower, buildings heat up naturally, so the company can reduce heat delivery without impacting the internal climate. Then, as people head to work, heat delivery can be increased to keep building temperatures stable.

Optimizing production across the system has brought about benefits both for customers and the environment. By managing the delivery of heat, Stockholm Exergi has reduced overall energy consumption, which not only saves costs but also limits the burning of fossil fuels and CO2 emissions. Since using Intelligy Solutions on AWS, Stockholm Exergi claims that energy consumption per customer has decreased by 5–10 percent. This amounts to an estimated 18,000–36,000 tons of CO2 emissions avoided annually. Customer costs have also decreased by 5–10 percent.

Expanding Responsible Energy Consumption with a Software-as-a-Service Solution

Stockholm Exergi is making Intelligy Solutions into a software-as-a-service solution on AWS that can be quickly adopted by other sustainability-minded energy providers. “We’re proud of our progress,” says Ulf Wikstrom, sustainability manager at Stockholm Exergi. “We’ve reduced our carbon footprint per square meter of building area from 20 kg of CO2 equivalent to less than 4 kg of CO2 equivalent.”

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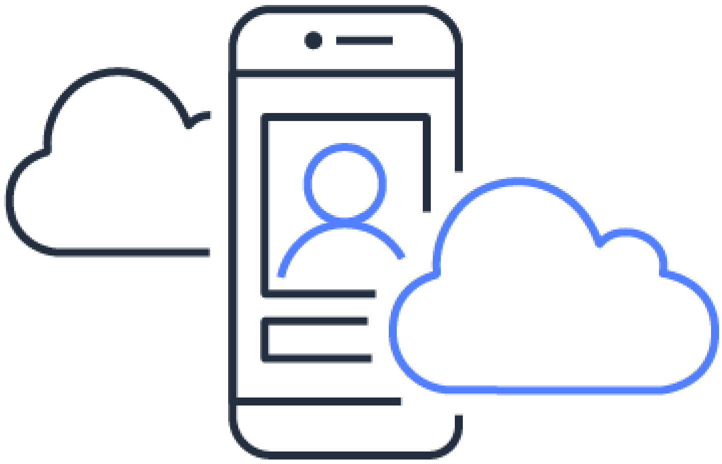

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2021

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